

- Q.30 Differentiate between mechanical pulping and chemical pulping.
- Q.31 Explain in brief the saponification process.
- Q.32 Draw a flow sheet for manufacturing of ethanol by fermentation of molasses.
- Q.33 Describe the major engineering problems associated with production of cement in brief.
- Q.34 Explain in brief block diagram process flow sheet.
- Q.35 Draw a flow sheet for extraction of vegetable oils by mechanical method.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the manufacturing process of ammonia along with flow sheet, uses and major engineering problems.
- Q.37 Explain the manufacturing process of sugar from sugarcane along with the flow sheet. Also discuss the major engineering problems associated with it.
- Q.38 Write short note on any two of the following
- Kraft process
 - Continuous hydrolysis
 - NPK fertilizer
 - Screen separators.

No. of Printed Pages : 4

180543/120543/030552

Roll No.

/564/030565/030562/544

4th Sem / Chem Subject:- Chemical Technology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 In paper making which of the following is used
- Packed bed
 - Distillation column
 - Fourdrinier machine
 - None of the above
- Q.2 Which of the following is a size reduction equipment?
- Jaw crusher
 - Cyclone separator
 - Tray drier
 - Packed tower
- Q.3 Vegetable oils and fats basically differ in their
- Physical state
 - Density ranges
 - Chemical Composition
 - Chemical properties
- Q.4 Cyclone separator is working on the principle of
- Density difference
 - Centrifugal force
 - Both (a) and (b)
 - None of the above
- Q.5 Urea is a
- Potassic fertilizer
 - Phosphate fertilizer
 - Nitrogenous fertilizer
 - None of the above

(100)

(4) 180543/120543/030552
/564/030565/030562/544

(1) 180543/120543/030552
/564/030565/030562/544

- Q.6 Gypsum is added to clinker during cement manufacturing to
- Decrease the rate of setting of cement
 - Make the cement impervious
 - Bind the particles of calcium silicate
 - To facilitate the formation of colloidal gel.
- Q.7 Which of the following tells the amount of free fatty acids presents in fat?
- Iodine number
 - Saponification number
 - Acid number
 - Richert-Meissl number
- Q.8 Chemical formula of oleum is
- H_2SO_4
 - H_2SO_3
 - $\text{H}_2\text{S}_2\text{O}_6$
 - $\text{H}_2\text{S}_2\text{O}_7$
- Q.9 Kraft process is related to which industry
- Fertilizer industry
 - Cement industry
 - Sugar industry
 - Pulp and paper industry
- Q.10 Soap are _____ based soapy detergents.
- Water
 - Kerosene
 - Oil
 - Acid

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define Pulp.
- Q.12 Draw the symbol used for heat exchanger.

(2) 180543/120543/030552
/564/030565/030562/544

- Q.13 Define fermentation.
- Q.14 Write one use of urea.
- Q.15 Name any two type of pump used in chemical industry.
- Q.16 Name the process used for manufacturing of Portland cement.
- Q.17 Write the name of a fertilizer containing phosphorus.
- Q.18 Write the chemical formula of caustic soda.
- Q.19 Name the catalyst used in hydrogenation of oil.
- Q.20 _____ (Lignin/ hemicellulose) increases the strength of the paper.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain in brief piping and instrumentation diagram.
- Q.22 Differentiate between soaps and detergents.
- Q.23 Discuss in brief the major engineering problems related to paper and pulp industry.
- Q.24 Explain in brief manufacturing of starch from maize.
- Q.25 Write the properties and uses of Ammonium Nitrate.
- Q.26 Discuss the major engineering problems related to manufacturing of triple super phosphate.
- Q.27 Draw a neat and clean flow sheet of manufacturing process of urea.
- Q.28 Discuss the properties and various grades of sulfuric acid.
- Q.29 Explain in brief the different types of cement.

(3) 180543/120543/030552
/564/030565/030562/544